a. Procedure-specific SSI incidence and burden

Procedure	Operations	SSI cases	Observed / predicted (95% CI)	Annual trend aIRR; q
Bile, liver, pancreas	80,789	2,324	1.01 (0.93-1.14)	1.027; =0.118
Liver transplant	9,878	263	0.55 (0.46-0.69)	NE
Colon	295,932	6,725	0.89 (0.83-0.93)	0.993; =0.500
Small bowel	245,423	4,098	0.84 (0.77-0.90)	1.024; =0.037
Spinal fusion	410,369	3,036	0.91 (0.83-0.99)	1.046; <0.001
Heart transplant	4,671	32	0.56 (0.22-0.95)	NE
Fracture reduction	400,655	2,697	0.94 (0.88-1.00)	1.018; =0.145
Spleen	18,043	120	1.09 (0.87-1.33)	0.974; =0.500
Kidney transplant	25,247	155	0.96 (0.68-1.32)	1.078; =0.118
Abdominal hysterectomy	204,922	1,253	0.83 (0.76-0.90)	1.017; =0.381
Rectal	64,016	391	0.39 (0.29-0.54)	0.953; =0.110
Exploratory laparotomy	452,467	2,699	1.02 (0.94-1.11)	1.027; =0.110
Coronary bypass, chest + donor	131,623	783	0.80 (0.70-0.92)	1.007; =0.776
Hip prosthesis	415,988	2,399	0.91 (0.83-0.98)	1.013; =0.436
Gastric	202,535	1,026	0.69 (0.63-0.74)	1.015; =0.500
Coronary bypass, chest only	16,877	82	0.70 (0.51-0.92)	0.996; =1.000
Vaginal hysterectomy	42,259	196	0.88 (0.73-1.07)	1.024; =0.505
Aortic aneurysm repair	3,487	16	0.67 (0.37-1.11)	NE
Knee prosthesis	496,006	1,733	0.92 (0.84-1.01)	1.015; =0.436
Cardiac	156,011	543	0.76 (0.61-0.90)	0.990; =0.776
Kidney	73,759	248	0.99 (0.81-1.17)	1.026; =0.436
Gallbladder	507,649	1,705	0.88 (0.81-0.95)	1.013; =0.436
Thoracic	206,270	588	0.79 (0.66-0.91)	1.061; =0.032
Laminectomy	342,895	941	0.79 (0.70-0.88)	0.981; =0.381
Appendix	378,956	1,001	0.68 (0.61-0.76)	0.937; <0.001
Pacemaker	172,140	318	1.06 (0.87-1.25)	0.975; =0.467
Cesarean	1,200,547	2,029	0.92 (0.83-1.03)	1.054; =0.004
Ovarian	178,558	124	0.95 (0.77-1.13)	0.973; =0.505



b. SSI burden by current and preceding annual reports

Population or procedure	Operations	SSIs	Next-year signal aRR (95% CI)	P / BH q
All ages	6,737,972	37,525	1.78 (1.70-1.87)	<0.001
Adult	6,526,888	36,470	NE	NE
Pediatric	211,084	1,055	NE	NE
Bile, liver, pancreas	80,789	2,324	2.57 (2.06-3.21)	<0.001
Liver transplant	9,878	263	NE	NE
Colon	295,932	6,725	1.76 (1.53-2.02)	<0.001
Small bowel	245,423	4,098	1.69 (1.48-1.93)	<0.001
Spinal fusion	410,369	3,036	2.10 (1.78-2.48)	<0.001
Heart transplant	4,671	32	NE	NE
Fracture reduction	400,655	2,697	1.51 (1.32-1.73)	<0.001
Spleen	18,043	120	2.11 (1.24-3.58)	=0.008
Kidney transplant	25,247	155	1.81 (1.09-3.01)	=0.030
Abdominal hysterectomy	204,922	1,253	1.58 (1.34-1.86)	<0.001
Rectal	64,016	391	1.55 (0.96-2.49)	=0.089
Exploratory laparotomy	452,467	2,699	1.69 (1.48-1.92)	<0.001
Coronary bypass, chest + donor	131,623	783	1.61 (1.27-2.03)	<0.001
Hip prosthesis	415,988	2,399	1.47 (1.28-1.68)	<0.001
Gastric	202,535	1,026	1.55 (1.26-1.91)	<0.001
Coronary bypass, chest only	16,877	82	1.08 (0.53-2.20)	=0.931
Vaginal hysterectomy	42,259	196	1.50 (1.02-2.23)	=0.055
Aortic aneurysm repair	3,487	16	NE	NE
Knee prosthesis	496,006	1,733	1.87 (1.63-2.16)	<0.001
Cardiac	156,011	543	1.72 (1.33-2.23)	<0.001
Kidney	73,759	248	1.31 (0.85-2.02)	=0.274
Gallbladder	507,649	1,705	1.57 (1.35-1.82)	<0.001
Thoracic	206,270	588	2.09 (1.63-2.69)	<0.001
Laminectomy	342,895	941	1.83 (1.50-2.22)	<0.001
Appendix	378,956	1,001	1.54 (1.29-1.84)	<0.001
Pacemaker	172,140	318	2.83 (2.07-3.87)	<0.001
Cesarean	1,200,547	2,029	1.96 (1.66-2.30)	<0.001
Ovarian	178,558	124	1.11 (0.49-2.49)	=0.931



- Current report at/below predicted
- Above after prior at/below
- Above in two consecutive years
- Above with prior year unavailable